JAIRAJ JOSHI

Senior Mechanical Engineer | Product Development, Manufacturing & Systems Integration

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PROFESSIONAL SUMMARY

Results-driven Senior Mechanical Engineer with 6+ years of experience leading product development, manufacturing process optimization, and electromechanical systems integration across highly regulated and commercial environments. Proven record delivering $370K+ in cost savings, 40% failure rate reductions, and accelerated time-to-market through rigorous design verification, DFM/DFA, and root cause analysis. Deep expertise spanning mechanical component design, fluid and thermal systems, injection molding, sheet metal, machining, and precision assembly. Skilled in SolidWorks, PTC Creo/Pro-E, Windchill PLM, DFMEA/PFMEA, DOE, GD&T, and Lean/Six Sigma methodologies. Experienced cross-functional leader and mentor, adept at coordinating with global suppliers, manufacturing teams, and engineering stakeholders to deliver high-quality, cost-effective solutions on schedule.

CORE COMPETENCIES

Mechanical & Electromechanical Design • Product Lifecycle Management (Concept to Launch) • DFM / DFA / DFMEA / PFMEA • Manufacturing Process Optimization & Validation (IQ/OQ/PQ) • CAD: SolidWorks / PTC Creo-Pro-E / AutoCAD • PLM: Agile / Windchill / SAP / Teamcenter • GD&T / Tolerance Stack-Up Analysis (ASME Y14.5) • Thermodynamics & Fluid Dynamics • Injection Molding / Sheet Metal / Machining • Root Cause Analysis (8D, DMAIC, 5-Why, Fishbone) • DOE / Gage R&R / SPC / Minitab • Lean Manufacturing & Six Sigma (Yellow Belt) • Supplier Qualification & Global Supplier Coordination • Cross-Functional Leadership & Team Mentorship • Regulatory: FDA 21 CFR Part 820 / ISO 13485 / IEC 60601-1

PROFESSIONAL EXPERIENCE

Sr. Sustaining R&D Engineer II – Catheter-Based Systems | Teleflex, Maple Grove, MN

January 2022 – June 2025

Led end-to-end design improvement, manufacturing process optimization, and product lifecycle management for Class III fluid-handling catheter systems. Managed phase gate reviews, design documentation (DHF, DFMEA, BOMs), and cross-functional collaboration with manufacturing, quality, and global supplier teams using SAP, Agile PLM, and SolidWorks.

- Led end-to-end design improvement projects for Class III catheter fluid-handling systems by developing project plans, managing phase gate reviews, and presenting design decisions to leadership, resulting in consistent on-time delivery across 3–5 concurrent product lines with zero regulatory nonconformances.
- Delivered $250K+ in annual COGS savings by engineering DFM-driven part consolidation, lean material substitutions, injection molding and sheet metal process improvements, and supplier-led tooling changes across catheter-based assemblies.
- Reduced field failure rates by 40% across 3–5 product lines by applying systematic root cause analysis (8D, 5-Why, Fishbone, DMAIC) to identify failure mechanisms and implement corrective design and process actions.
- Coordinated product development activities with 8+ overseas suppliers across 3 countries, achieving 98% milestone adherence and maintaining engineering quality standards throughout global supply chain.
- Executed comprehensive manufacturing process validations (IQ/OQ/PQ) and Gage R&R studies, achieving consistently >95% inspection accuracy and supporting FDA QSR and ISO 13485 compliance.
- Mentored 5+ junior engineers and led 50+ design reviews, achieving 95% on-time project delivery and accelerating project timelines by [~20%] through structured coaching and design input optimization.

Research & Development Engineer – Structural Heart Implants | Abbott Laboratories, Plymouth, MN

March 2021 – January 2022

Contributed to mechanical design, systems integration, and verification of Class III vascular occlusion devices and catheter delivery systems. Managed DFMEA/UFMEA, BOM control, and V&V documentation within Windchill PLM.

- Executed mechanical design verification for 8 Class III implantable systems using fatigue, tensile, and simulated-use protocols per ISO 10555 and ISO 5840, achieving 100% compliance and zero test failures across all submissions.
- Designed and validated 15+ custom test fixtures using SolidWorks and FEA, improving verification repeatability by 40% and reducing testing cycle times by 20%, enabling faster clinical feedback integration.
- Compressed design freeze timelines by 3–4 weeks by delivering rapid mechanical risk assessments through FEA-driven structured DFM reviews and updated PFMEA documentation, enabling same-sprint verification launch.
- Managed DFMEA/UFMEA updates for 12+ projects and controlled BOMs for 200+ components within Windchill PLM, achieving 99% documentation accuracy across clinical, regulatory, and R&D collaborations.
- Collaborated with physicians and clinical teams to translate use requirements and human factors inputs into mechanical design specifications, reducing post-verification issue rates by 15%.

Mechanical Sustaining Engineer (Consultant – Med Device Client) | Larsen & Toubro Technology Services, Cincinnati, OH

November 2020 – February 2021

Executed design validations, drawing remediations, tolerance stack-up analyses, and CAPA resolution for Class II breast biopsy devices under a regulatory remediation initiative using Agile MAP PLM.

- Reduced field complaint rates by 30% within 6 months by leading root cause investigations (8D, Fishbone) and implementing targeted mechanical design corrections, directly improving long-term product reliability.
- Achieved $120K in annual material savings by executing DFM-driven design modifications across 12 product variants while maintaining 100% clinical performance standards and regulatory compliance.
- Sustained 100% part availability for 3 critical components during global supply disruptions by executing compliant supplier requalification protocols including dimensional inspection, functional verification, and FDA QSR change documentation.

Engineer I – Infusion Systems (Disposable Devices) | Baxter International, Round Lake, IL

April 2019 – November 2020

Executed mechanical V&V, process validation, and risk management for sterile disposable fluid delivery systems under GMP, ISO 13485, and FDA 21 CFR 820, applying thermodynamics and fluid dynamics principles to optimize mechanical design.

- Reduced fluid system failure rates by 40% by designing custom verification fixtures and executing DOE-driven IQ/OQ/PQ validation protocols for tubing and connector assemblies, lowering field escape risk across 2 production lines.
- Improved manufacturing throughput by 15% by leading DFM-focused process optimization and executing structured tech transfer activities, reducing line changeover time and improving operator build consistency.
- Applied thermodynamics and fluid dynamics principles to optimize sterilization processes and mechanical integrity for sterile disposables, achieving 99% sterilization consistency and eliminating 100% of field-reported mechanical failures.
- Designed and validated 6 custom IQ/OQ test fixtures, improving mechanical integrity measurements and supporting first-cycle regulatory approval for 4 new product lines.

EDUCATION

Master of Science, Mechanical Engineering | Illinois Institute of Technology, Chicago, IL

Bachelor of Science, Mechanical Engineering | Nagpur University, India

CERTIFICATIONS & TRAINING

Lean Six Sigma Yellow Belt (EdX) • Google Project Management Certificate (In Progress) • Internal: Design Controls, DFMEA/PFMEA, GMP Compliance, DOE, Gage R&R, Team Mentorship & Performance Coaching